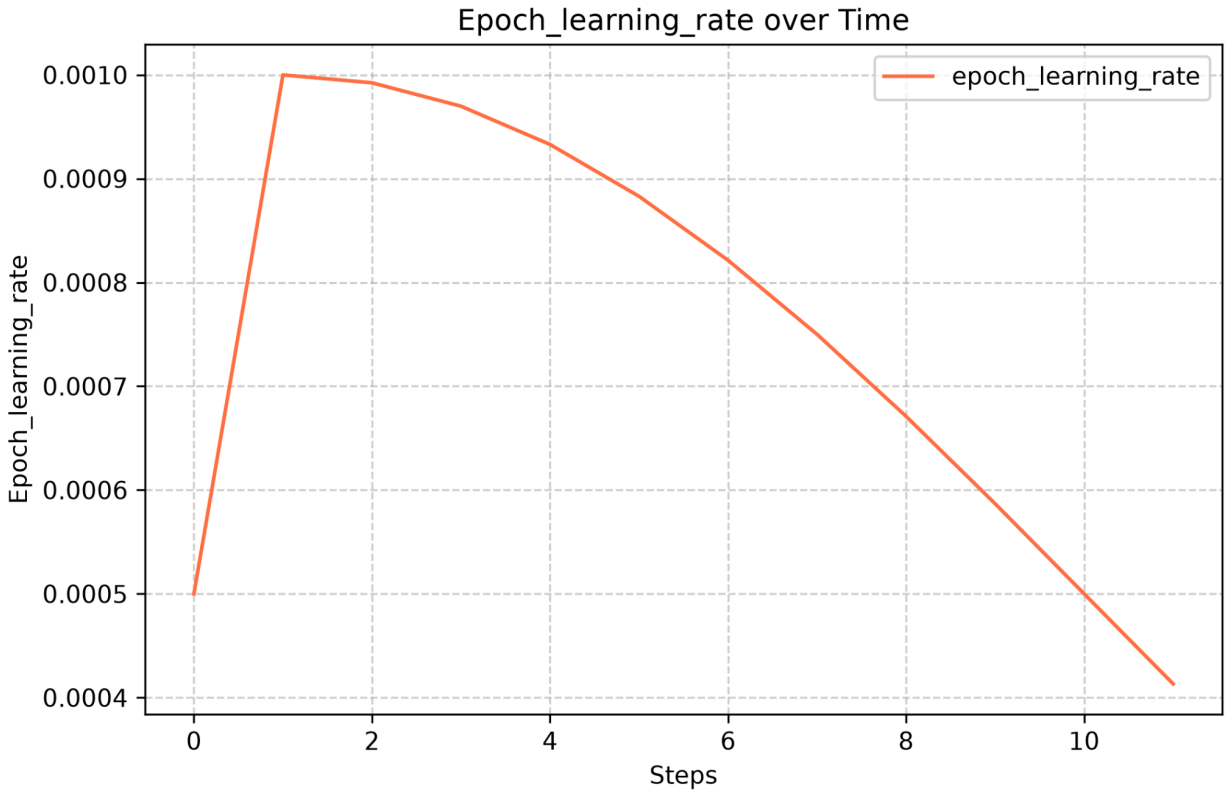
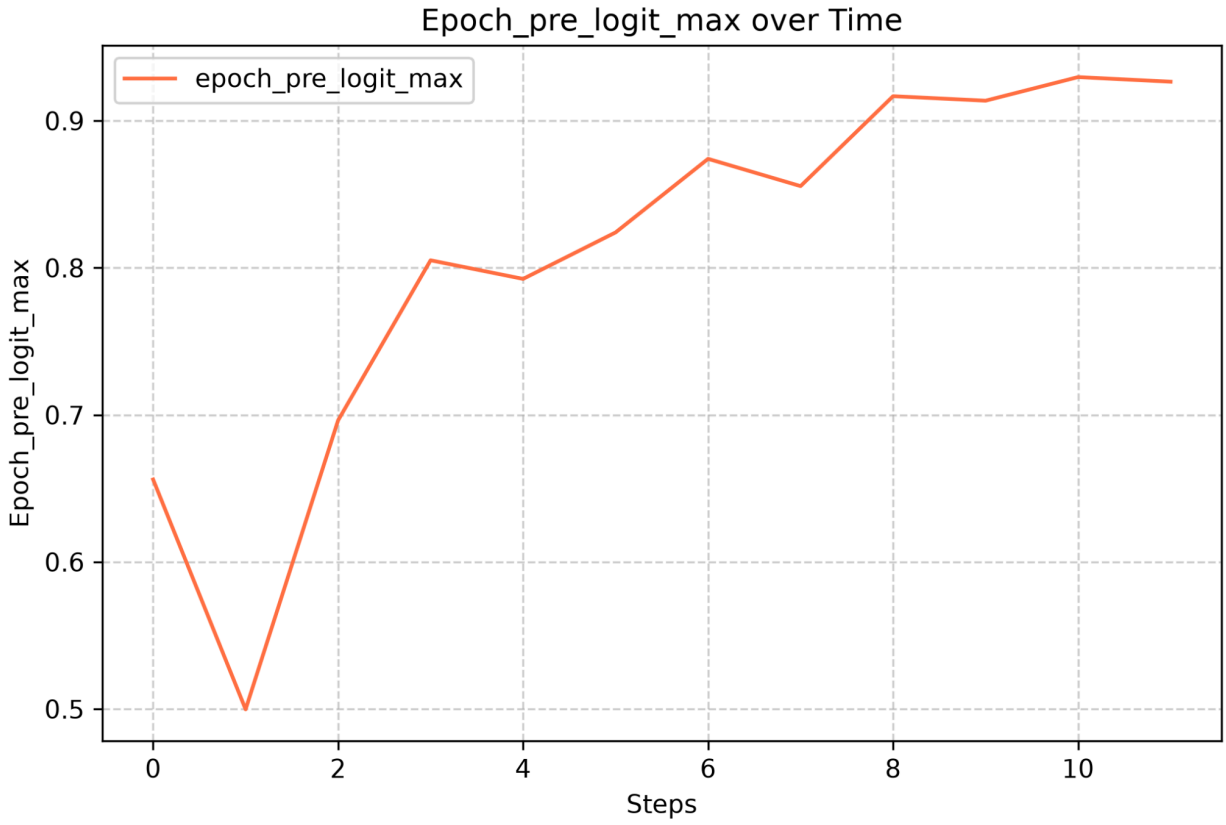
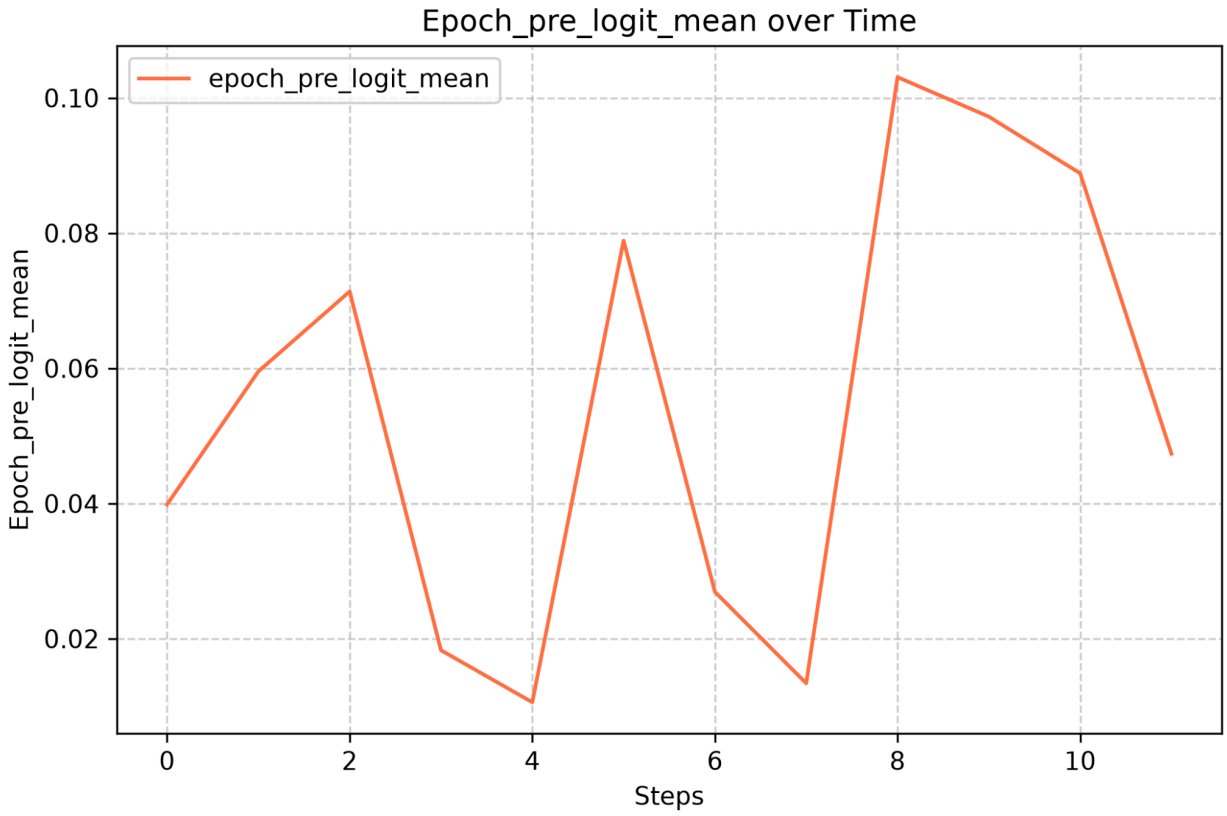
TMP75

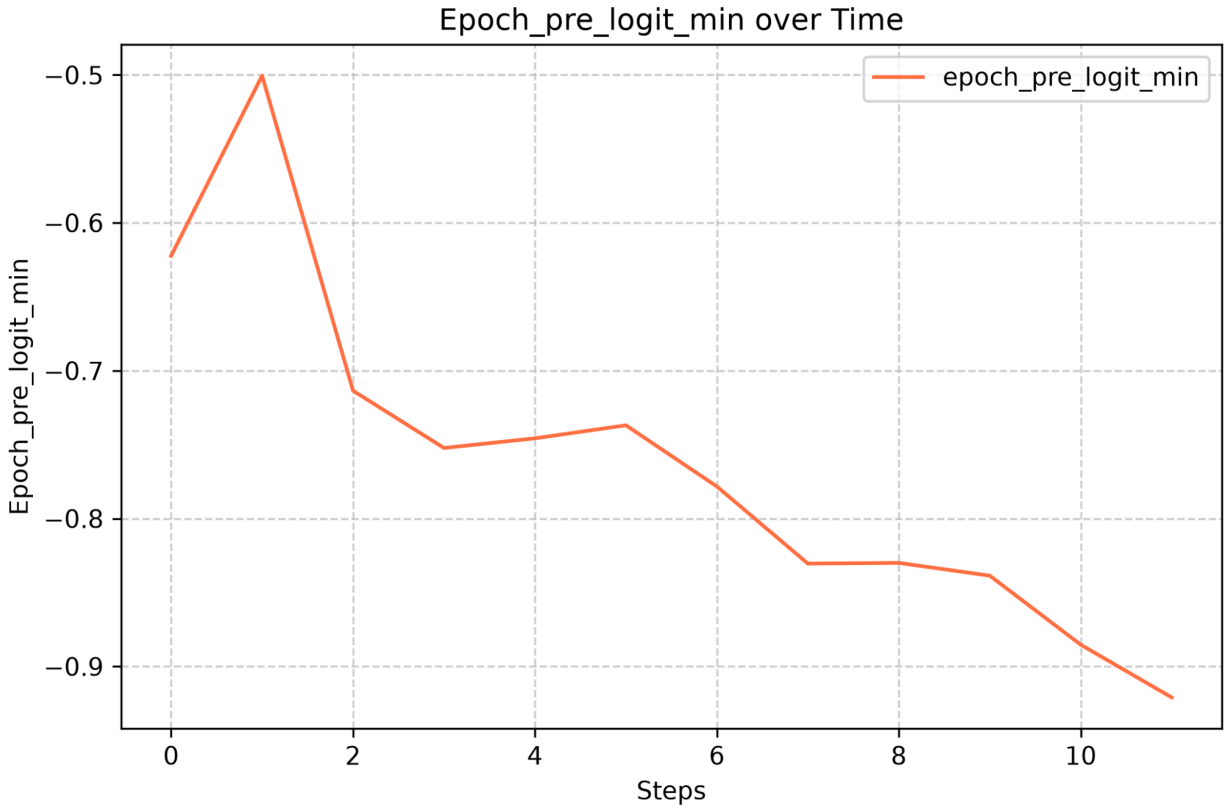
Fixed parameters:

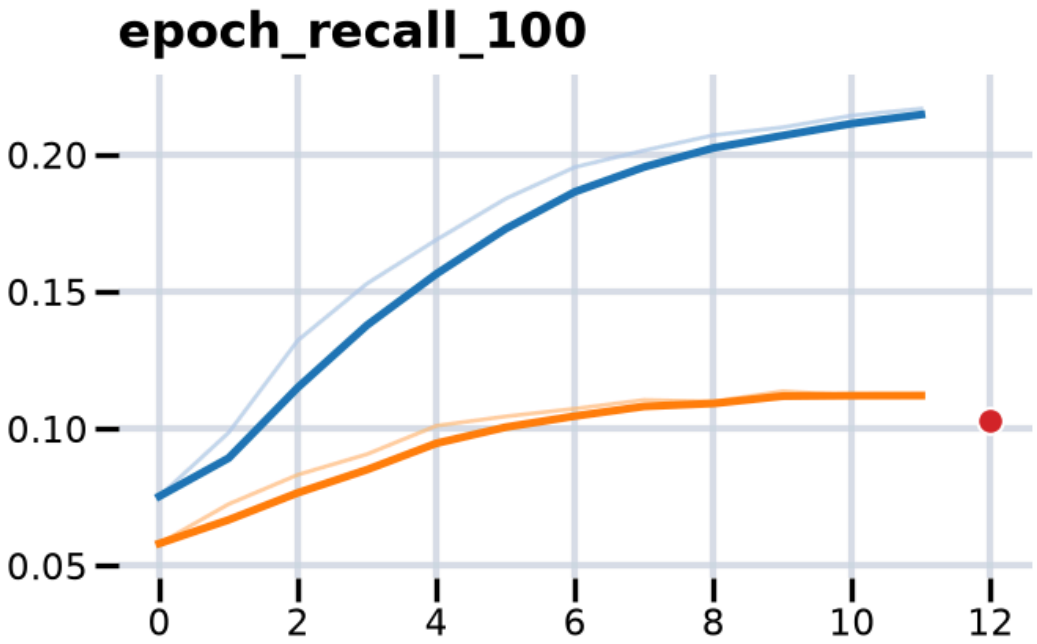
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"k": 100,
"learning_rate": 0.001,
"weight_decay": 0.02,
"drop_rate": 0.3,
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"regl2": 0.0,
"feature_acronym": "ahosy",
"incl_genres": true,
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"bias_corr_alpha": 0.05,
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"movie_tiers_uri":
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Epoch_learning_rate over Time
epoch_learning_rate
Epoch_learning_rate
0.0010
0.0009
0.0008
0.0007
0.0006
0.0005
0.0004
0
2
4
6
8
10
Steps

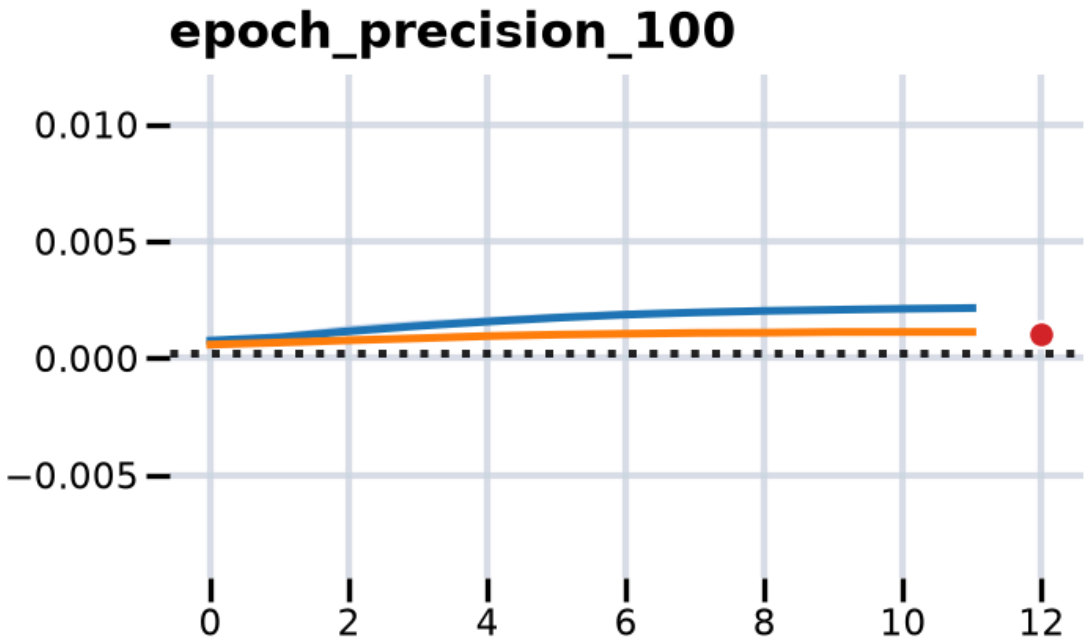

Epoch_pre_logit_max over Time
epoch_pre_logit_max
Epoch_pre_logit_max
0.9
0.8
0.7
0.6
0.5
0
2
4
6
8
10
Steps


Epoch_pre_logit_mean over Time
epoch_pre_logit_mean
Epoch_pre_logit_mean
0.10
0.08
0.06
0.04
0.02
0
2
4
6
8
10
Steps

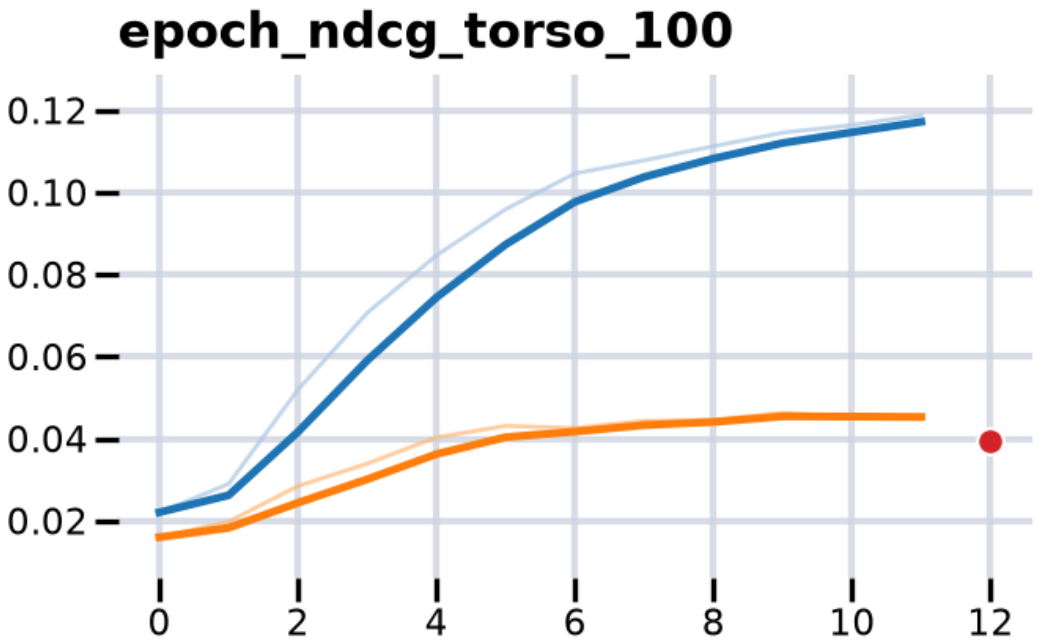

Epoch_pre_logit_min over Time
epoch_pre_logit_min
Epoch_pre_logit_min
−0.5
−0.6
−0.7
−0.8
−0.9
0
2
4
6
8
10
Steps



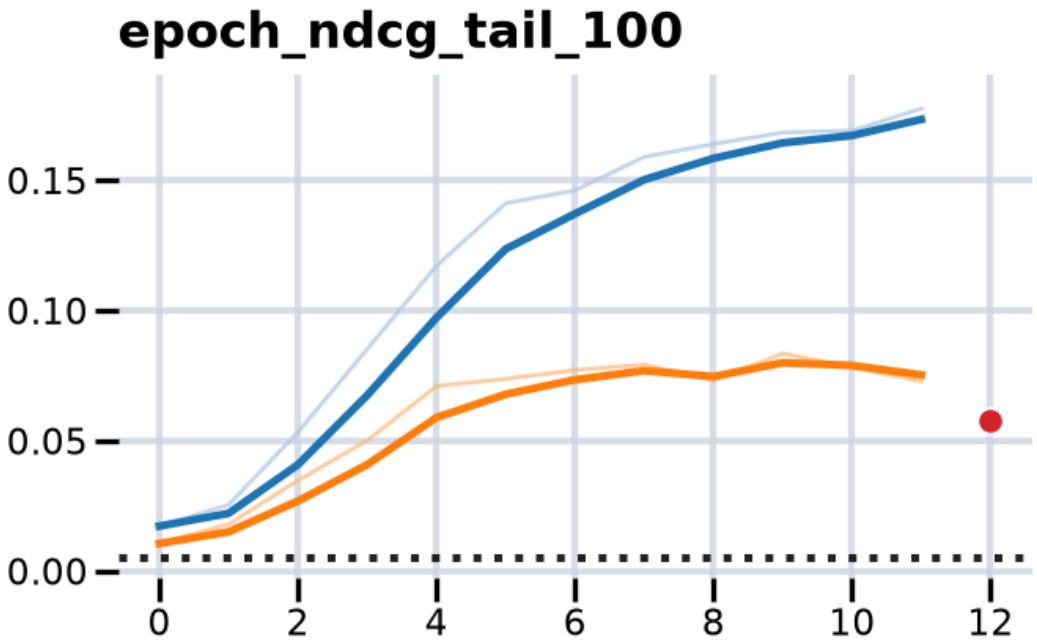
	Smoothed	Value	Step	Relative
train	0.2147	0.2169	11	11.43m
validation	0.1122	0.1122	11	11.43m
test	0.1030	0.1030	12	0.00m
ceiling	-	-	-	-
random	-	0.0258	-	-



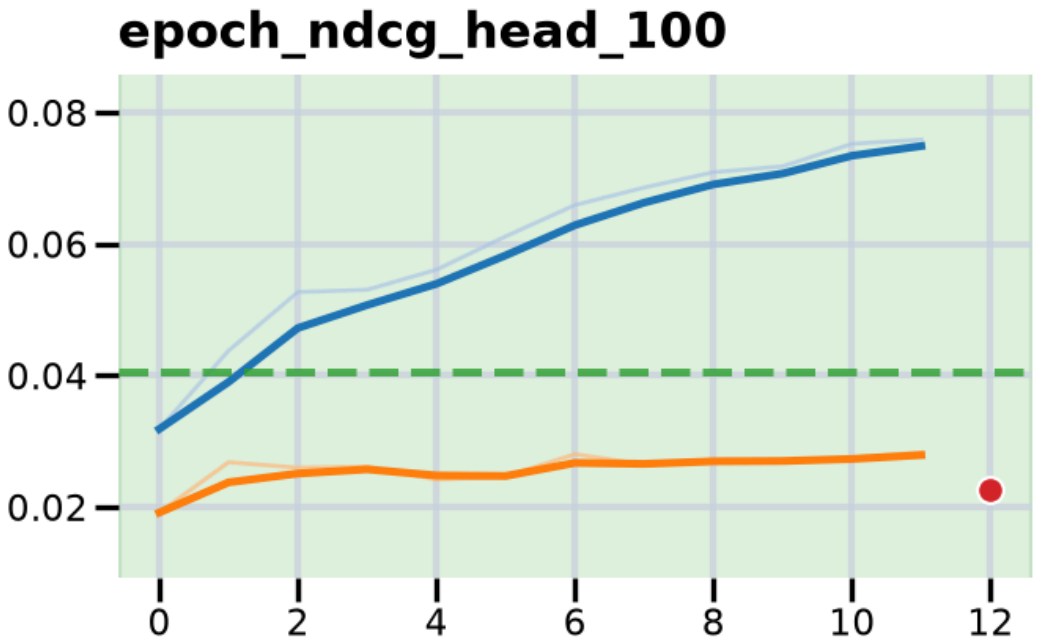
	Smoothed	Value	Step	Relative
train	0.0021	0.0022	11	11.43m
validation	0.0011	0.0011	11	11.43m
test	0.0010	0.0010	12	0.00m
ceiling	-	-	-	-
random	-	0.0003	-	-



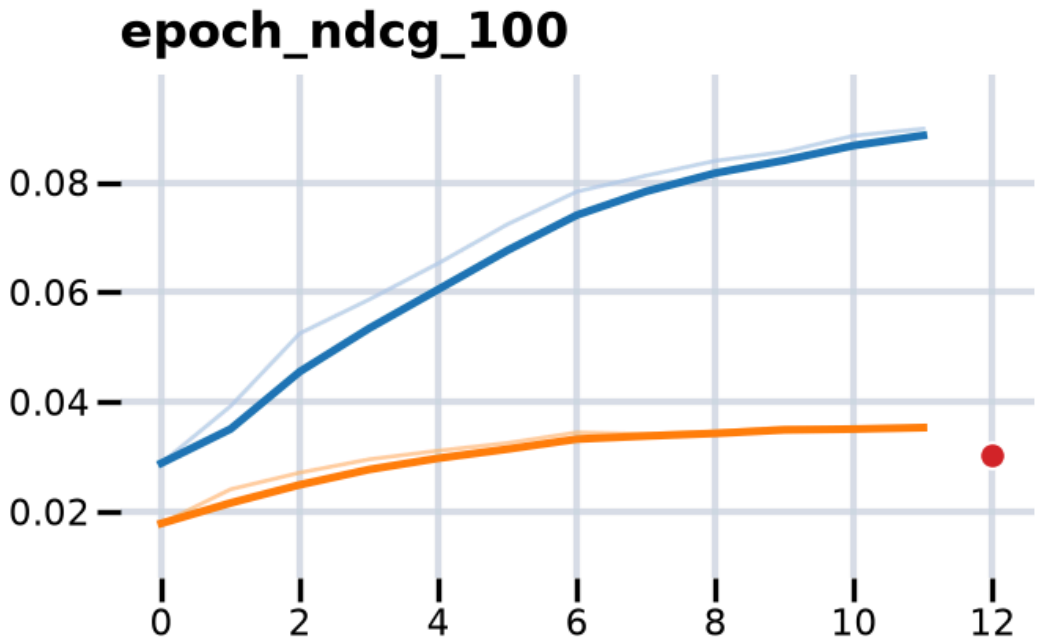
	Smoothed	Value	Step	Relative
train	0.1172	0.1189	11	11.43m
validation	0.0453	0.0452	11	11.43m
test	0.0395	0.0395	12	0.00m
ceiling	-	-	-	-
random	-	0.0054	-	-



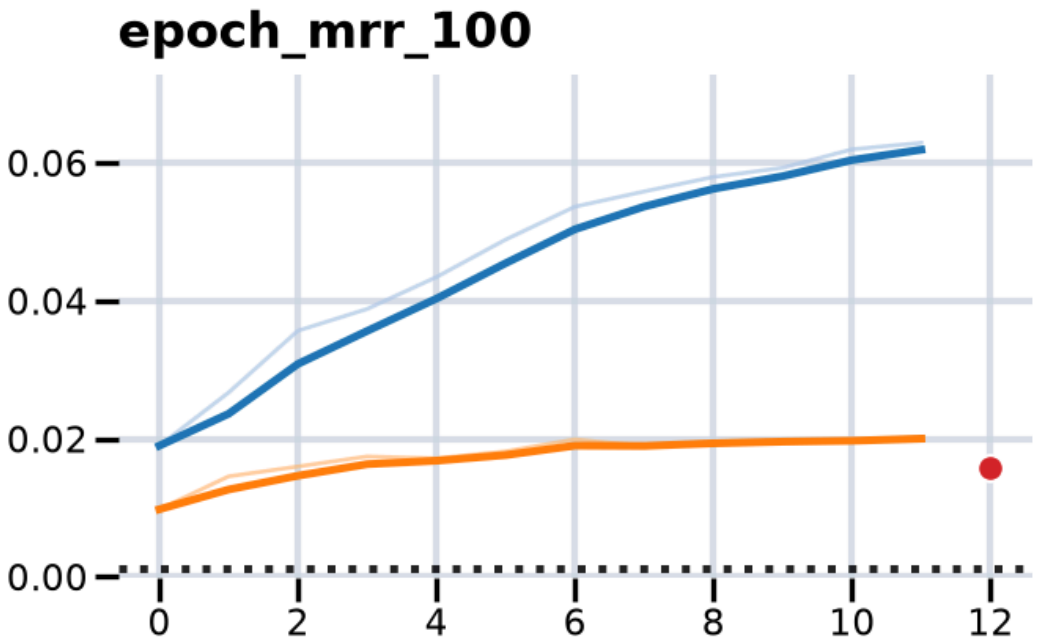
	Smoothed	Value	Step	Relative
train	0.1734	0.1775	11	11.43m
validation	0.0753	0.0729	11	11.43m
test	0.0581	0.0581	12	0.00m
ceiling	-	-	-	-
random	-	0.0054	-	-



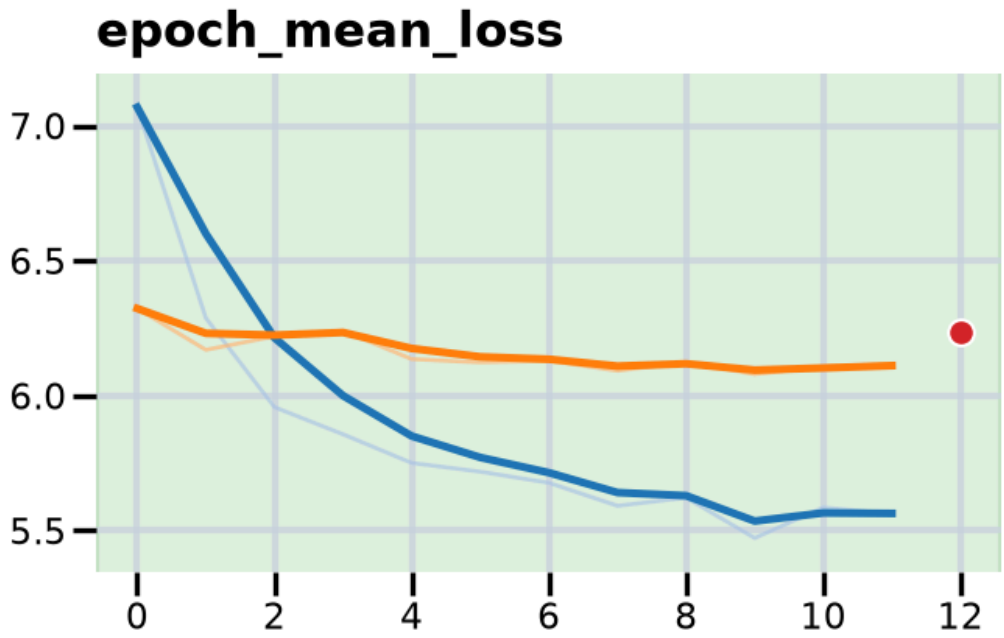
	Smoothed	Value	Step	Relative
train	0.0749	0.0759	11	11.43m
validation	0.0279	0.0283	11	11.43m
test	0.0227	0.0227	12	0.00m
ceiling	-	0.0405 ±1181.5	-	-
random	-	0.0054	-	-



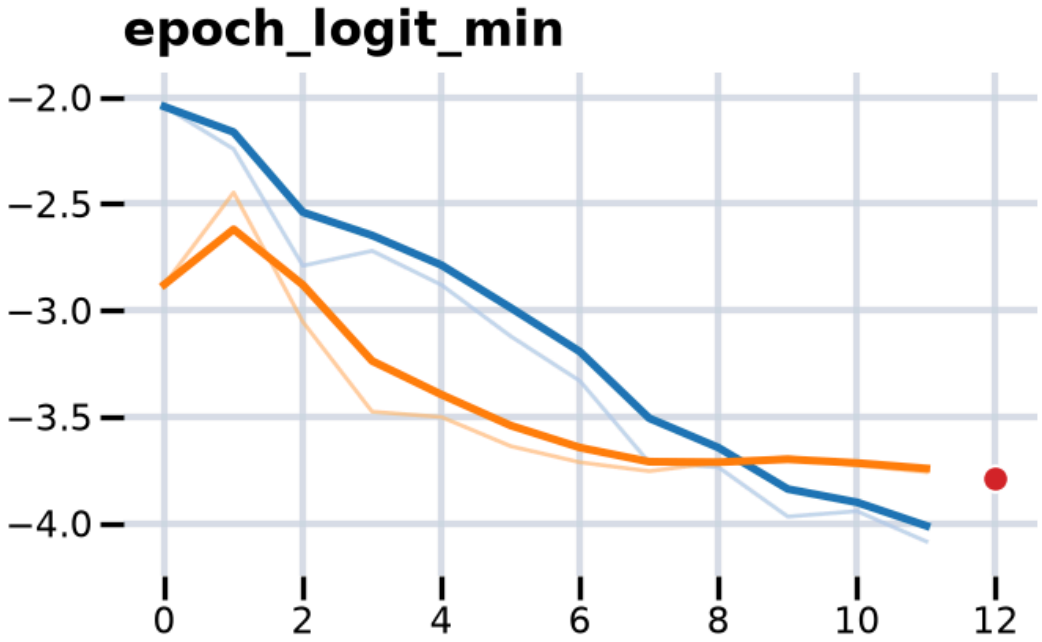
	Smoothed	Value	Step	Relative
train	0.0886	0.0898	11	11.43m
validation	0.0353	0.0355	11	11.43m
test	0.0303	0.0303	12	0.00m
ceiling	-	-	-	-
random	-	0.0054	-	-



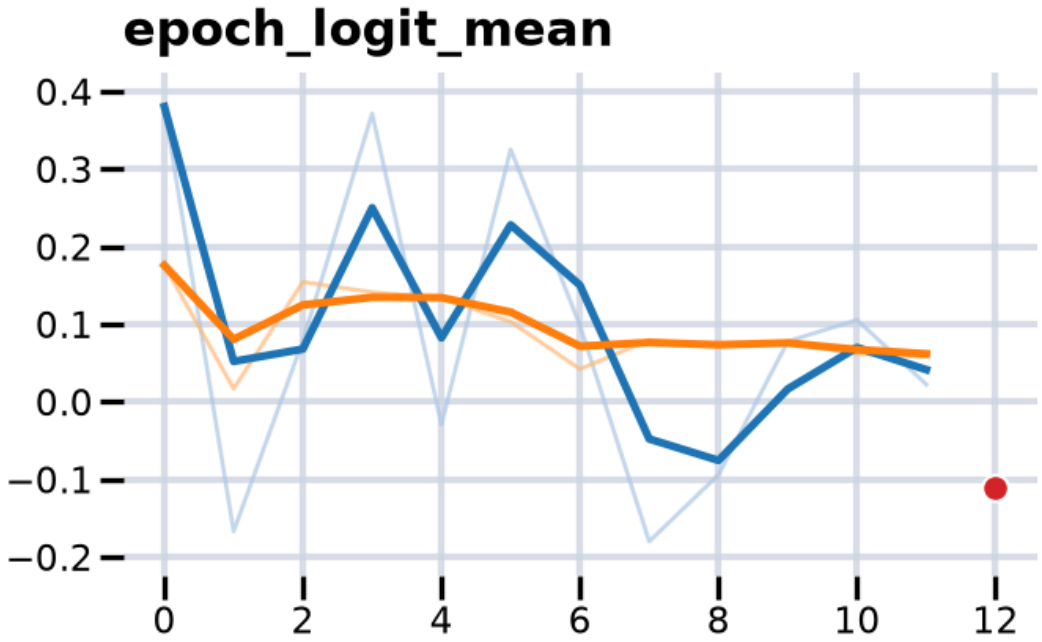
	Smoothed	Value	Step	Relative
train	0.0619	0.0629	11	11.43m
validation	0.0200	0.0202	11	11.43m
test	0.0158	0.0158	12	0.00m
ceiling	-	-	-	-
random	-	0.0013	-	-



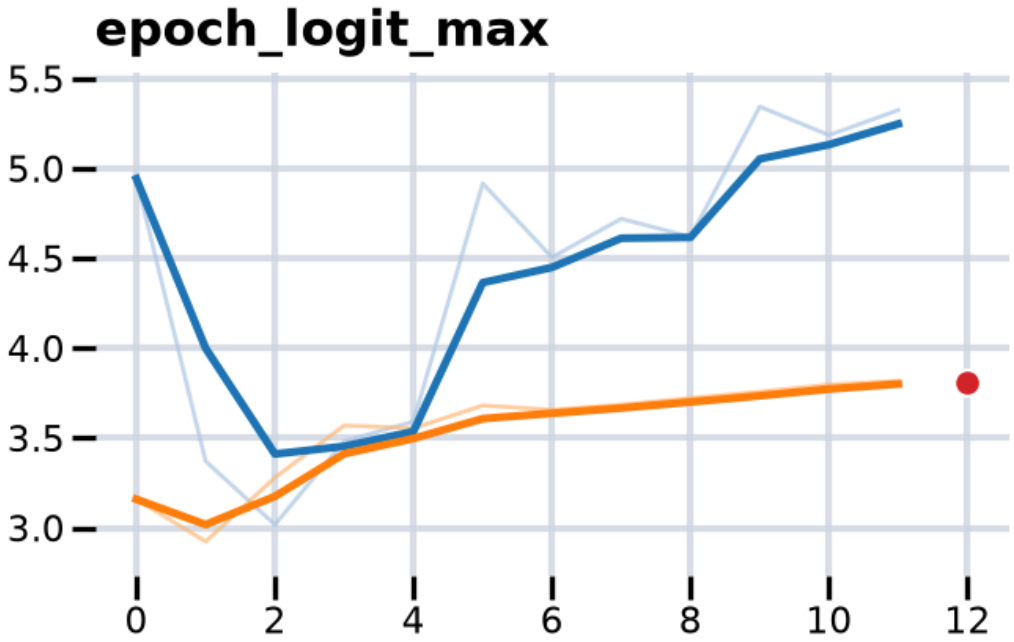
	Smoothed	Value	Step	Relative
train	5.5640	5.5629	11	11.43m
validation	6.1114	6.1171	11	11.43m
test	6.2383	6.2383	12	0.00m
irred error	-	0.0000 ±2725.51	-	-
random	-	7.6246	-	-



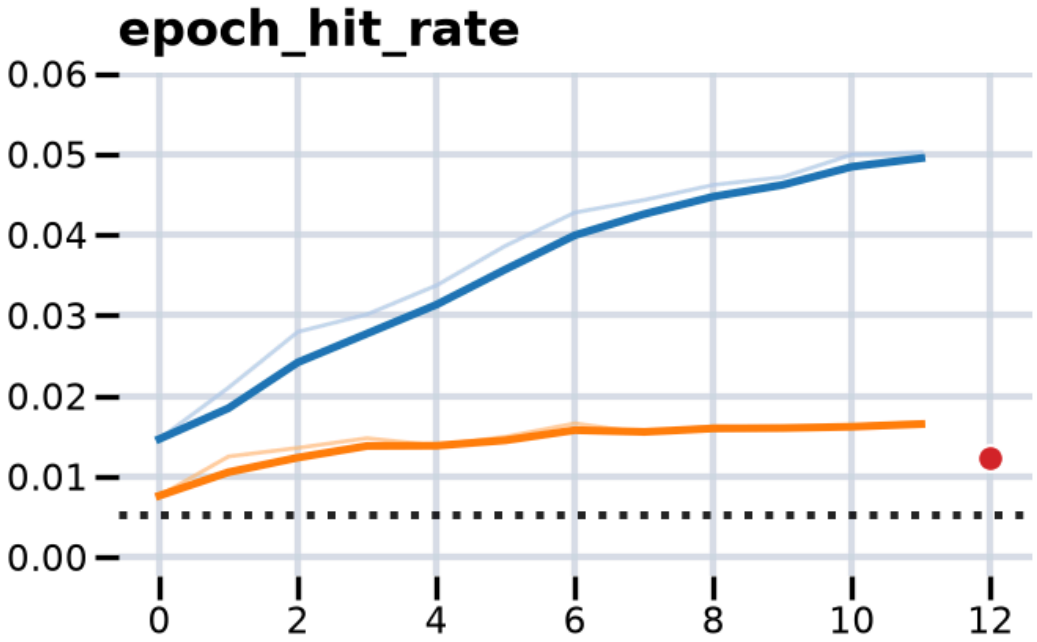
	Smoothed	Value	Step	Relative
train	-4.0117	-4.0853	11	11.43m
validation	-3.7425	-3.7597	11	11.43m
test	-3.7913	-3.7913	12	0.00m
ceiling	-	-	-	-



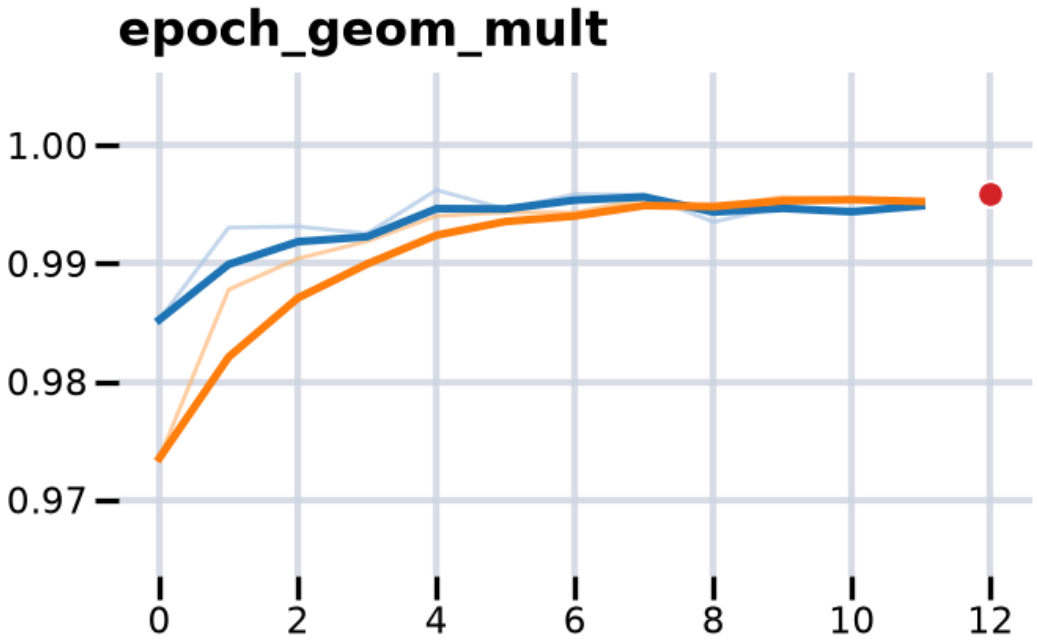
	Smoothed	Value	Step	Relative
train	0.0416	0.0226	11	11.43m
validation	0.0620	0.0584	11	11.43m
test	-0.1109	-0.1109	12	0.00m
ceiling	-	-	-	-



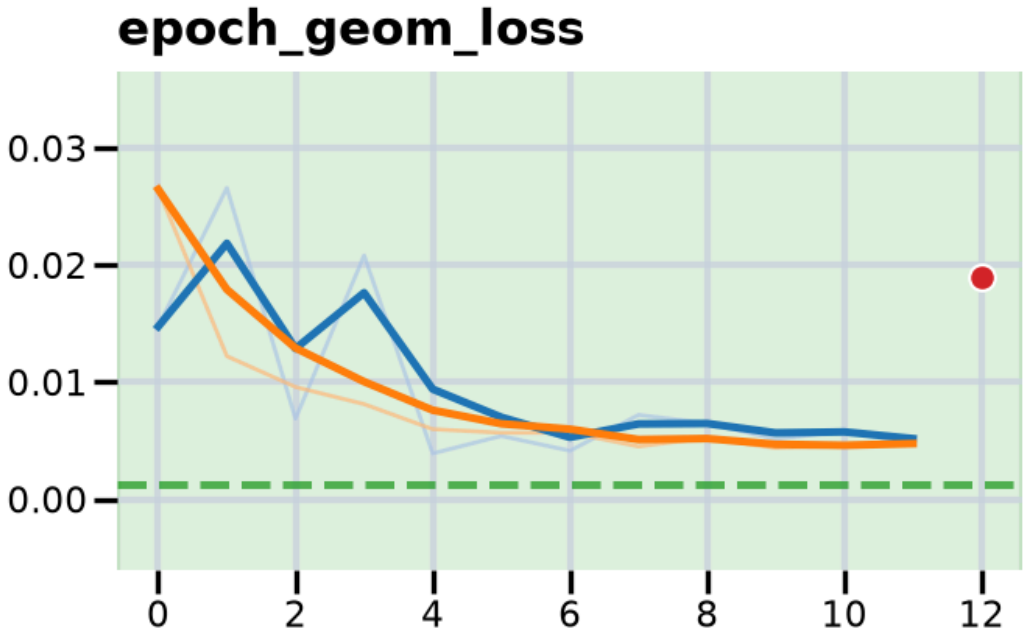
	Smoothed	Value	Step	Relative
train	5.2481	5.3249	11	11.43m
validation	3.8004	3.8192	11	11.43m
test	3.8102	3.8102	12	0.00m
ceiling	-	-	-	-



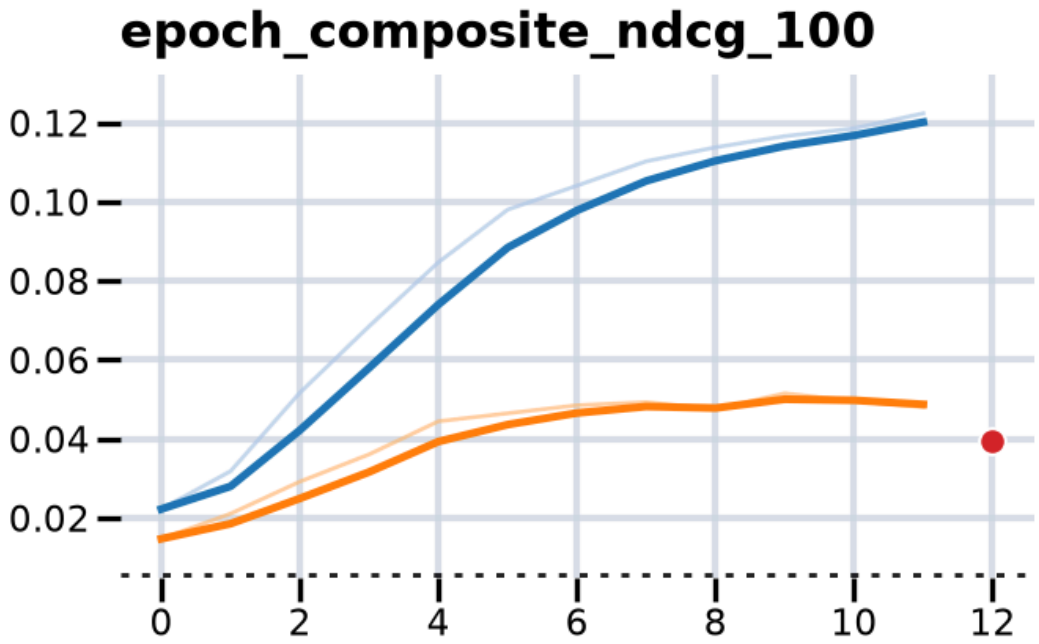
	Smoothed	Value	Step	Relative
train	0.0495	0.0502	11	11.43m
validation	0.0165	0.0167	11	11.43m
test	0.0124	0.0124	12	0.00m
ceiling	-	-	-	-
random	-	0.0054	-	-



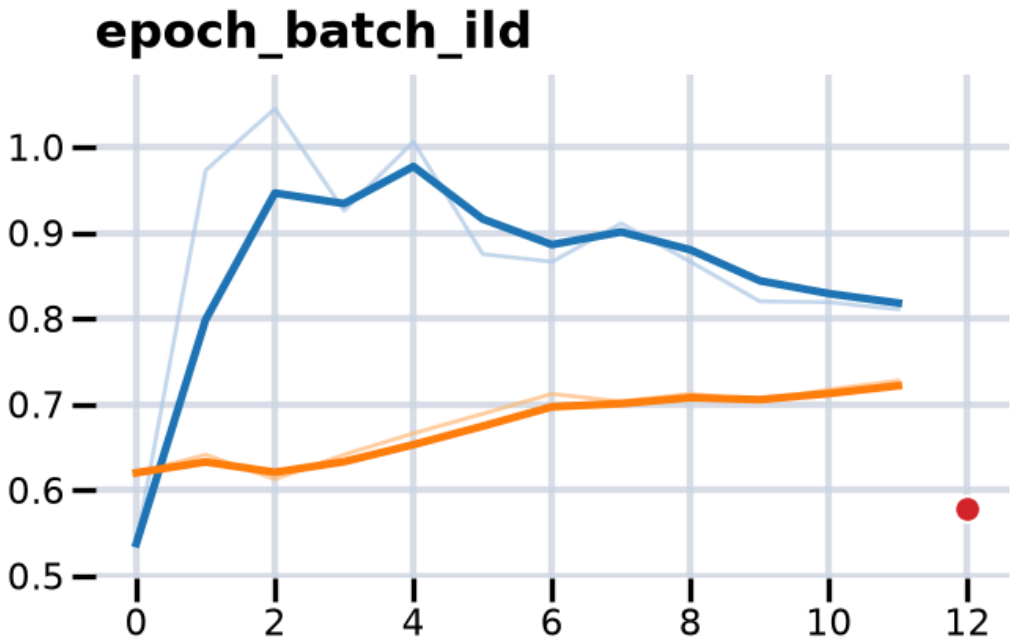
	Smoothed	Value	Step	Relative
train	0.9949	0.9952	11	11.43m
validation	0.9952	0.9951	11	11.43m
test	0.9959	0.9959	12	0.00m
ceiling	-	-	-	-



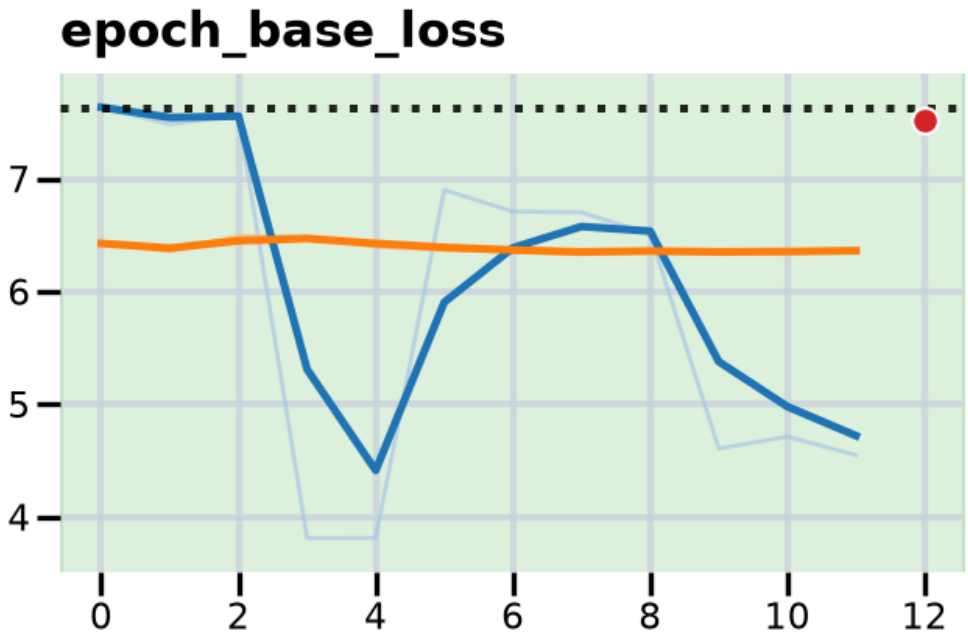
	Smoothed	Value	Step	Relative
train	0.0052	0.0048	11	11.43m
validation	0.0048	0.0049	11	11.43m
test	0.0189	0.0189	12	0.00m
irred error	-	0.0013 ±0.066	-	-
random	-	7.6246	-	-



	Smoothed	Value	Step	Relative
train	0.1202	0.1224	11	11.43m
validation	0.0488	0.0481	11	11.43m
test	0.0395	0.0395	12	0.00m
ceiling	-	-	-	-
random	-	0.0054	-	-

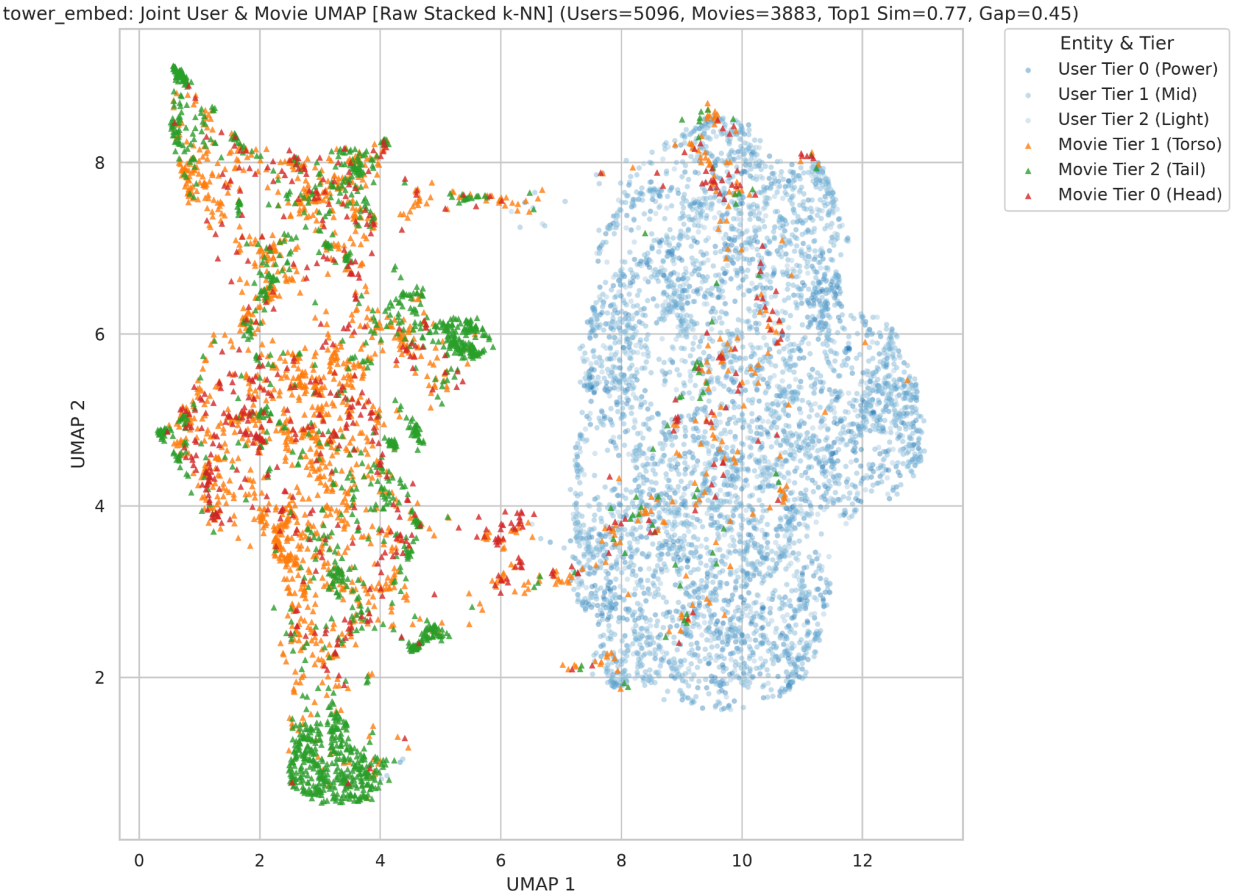


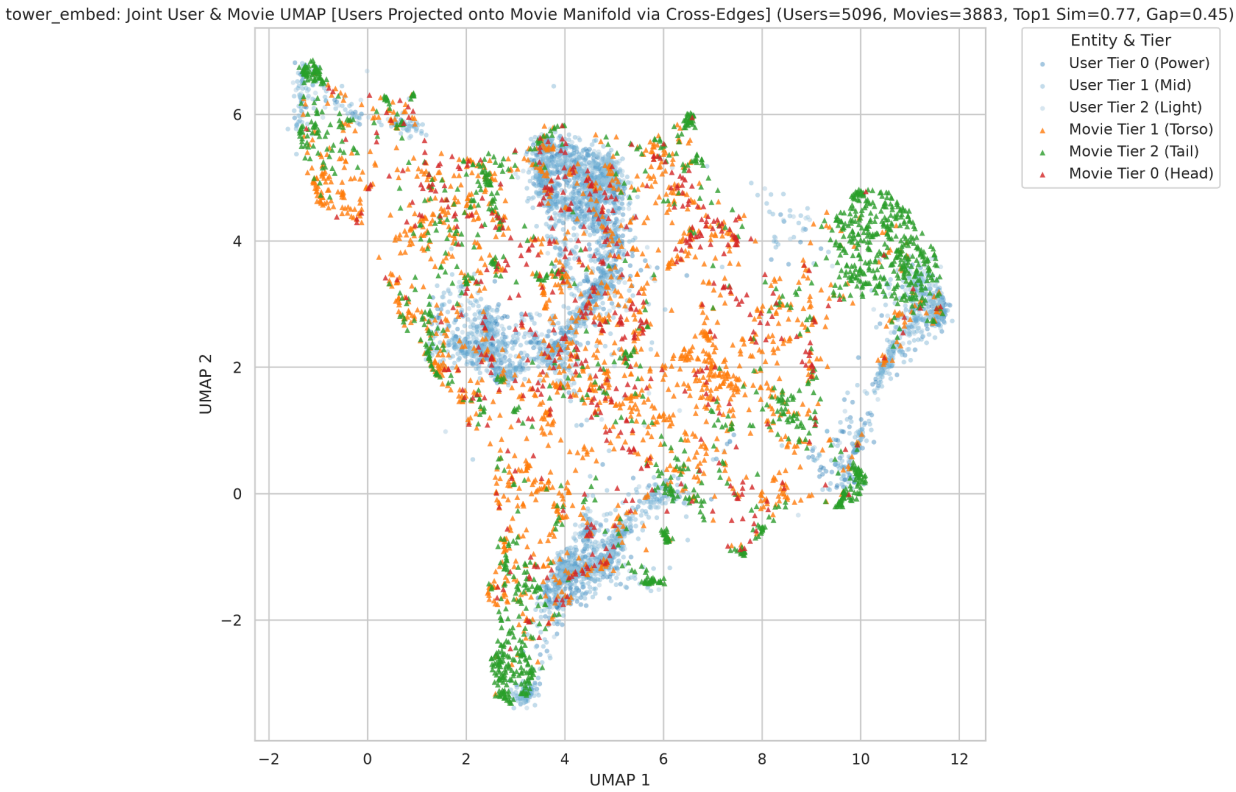
	Smoothed	Value	Step	Relative
train	0.8181	0.8108	11	11.43m
validation	0.7216	0.7277	11	11.43m
test	0.5788	0.5788	12	0.00m
ceiling	-	-	-	-



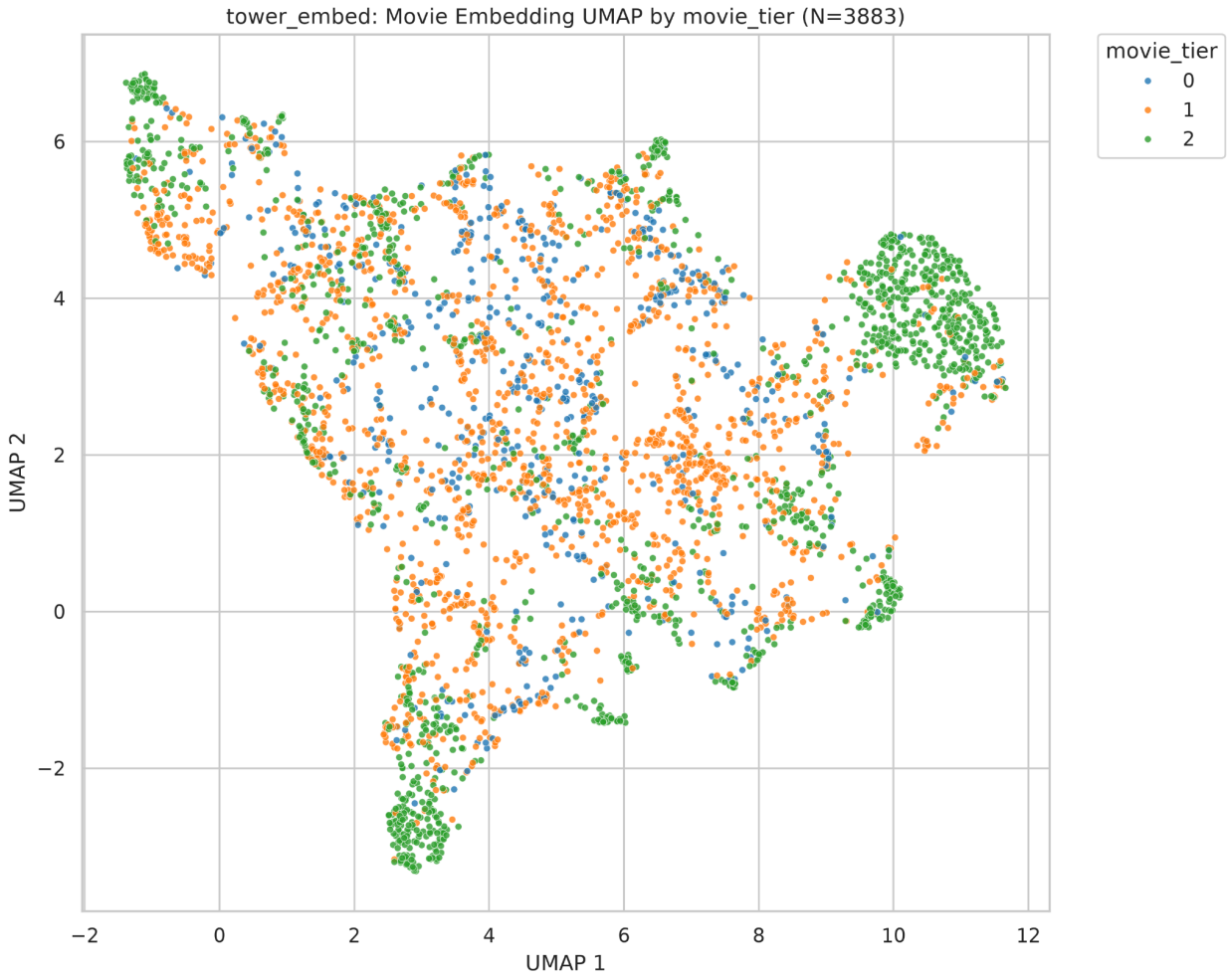
	Smoothed	Value	Step	Relative
train	4.7214	4.5495	11	11.43m
validation	6.3625	6.3682	11	11.43m
test	7.5221	7.5221	12	0.00m
irred error	-	0.0000 ±1042.63	-	-
random	-	7.6246	-	-

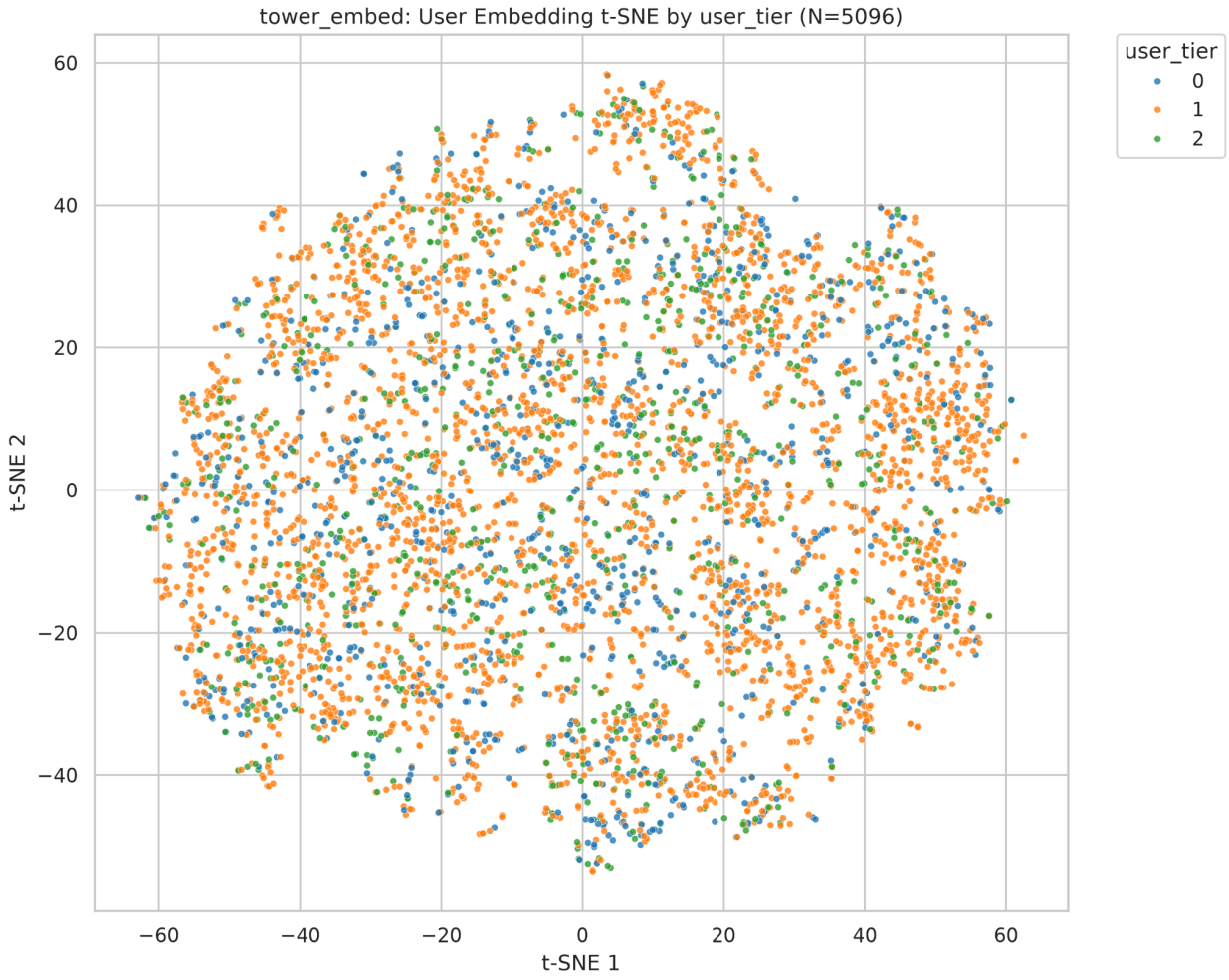

tower_embed: Joint User & Movie t-SNE [Modality-Centered Cosine t-SNE] (Users=5096, Movies=3883, Top1 Sim=0.77, Gap=0.45)
Entity & Tier
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
t-SNE 2
t-SNE 1
60
40
20
0
−20
−40
−60
−80
−60
−40
−20
0
20
40
60
80

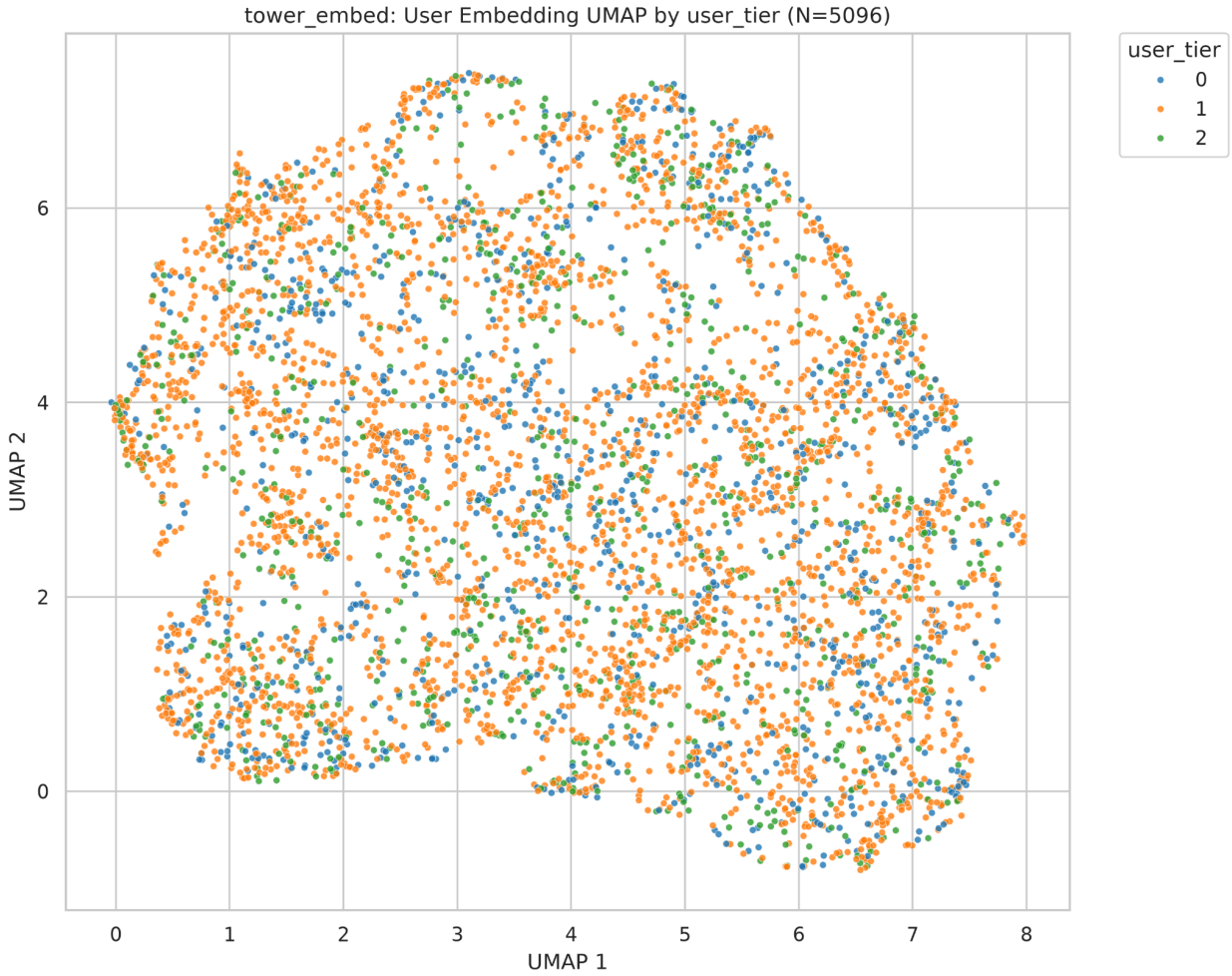

tower_embed: Joint User & Movie UMAP [Raw Stacked k-NN] (Users=5096, Movies=3883, Top1 Sim=0.77, Gap=0.45)
Entity & Tier
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
UMAP 2
UMAP 1
8
6
4
2
0
2
4
6
8
10
12


tower_embed: Joint User & Movie UMAP [Users Projected onto Movie Manifold via Cross-Edges] (Users=5096, Movies=3883, Top1 Sim=0.77, Gap=0.45)
Entity & Tier
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
UMAP 2
UMAP 1


tower_embed: Movie Embedding t-SNE by movie_tier (N=3883)
movie_tier
0
1
2
t-SNE 2
t-SNE 1


tower_embed: Movie Embedding UMAP by movie_tier (N=3883)
movie_tier
0
1
2
UMAP 2
UMAP 1
6
4
2
0
−2
−2
0
2
4
6
8
10
12


tower_embed: User Embedding t-SNE by user_tier (N=5096)
user_tier
0
1
2
t-SNE 2
t-SNE 1
60
40
20
0
−20
−40
−60
−40
−20
0
20
40
60



{
"coverage@100": [
"HEALTHY EXPOSURE BALANCE at_100: Gini is 0.72 (Effective Catalog: 1455 items, Normalized Entropy: 0.881). The model successfully balances mainstream relevance with catalog exploration.",
"STRONG TAIL at_100: The bottom 80% captures 26.7% of traffic, indicating excellent long-tail surfacing capability.",
"DIVERSE HEAD at_100: The top 10% consume 48.9% of slots, leaving plenty of room for the torso/tail.",
"LIGHT/NICHE USER DIVERSIFICATION at_100: Light/niche users (Tier 2 Equalized Gini: 0.723) receive more broadly distributed catalog exposure than power users (Tier 0 Equalized Gini: 0.755)."
],
"coverage@20": [
"HEALTHY EXPOSURE BALANCE at_20: Gini is 0.85 (Effective Catalog: 768 items, Normalized Entropy: 0.804). The model successfully balances mainstream relevance with catalog exploration.",
"MODERATE TAIL at_20: The bottom 80% captures 6.6% of traffic.",
"DIVERSE HEAD at_20: The top 10% consume 72.7% of slots, leaving plenty of room for the torso/tail.",
"UNIFORM COHORT CONCENTRATION at_20: Power (Tier 0: 0.877) and Light (Tier 2: 0.861) users experience roughly the same level of catalog concentration."

```
  ],
  "data_shifts": [
    "HIGH MOVIE TIER DRIFT WARNING: Significant item popularity shift between Train and Test splits (Train-Test EMD: 0.0679). Evaluation may be biased.",
    "USER TIER STABILITY EXCELLENT: Activity profile distribution remains consistent across splits (Train-Test EMD: 0.0113).",
    "EXPECTED INTERACTION POPULARITY BIAS: Catalog vs. Train EMD is 0.4699. Training interactions heavily concentrate on popular head (Tier 0) items relative to catalog availability.",
    "P(X) [Marginal Item Exposure] HIGH DRIFT WARNING: Significant shift between Train and Test splits (Train-Test EMD: 0.0932).",
    "P(Y=1|X) [Conditional Hit-Rate Profile] STABILITY EXCELLENT: Distribution remains consistent across splits (Train-Test EMD: 0.0054).",
    "P(Y=1) [Global Positive Rating Rate] HIGH DRIFT WARNING: Significant shift between Train and Test splits (Train-Test AbsDiff: 0.0761).",
    "P(X|Y=1) [Positive Item Tier Distribution] HIGH DRIFT WARNING: Significant shift between Train and Test splits (Train-Test EMD: 0.0679).",
    "Covariate shift between Train and Test splits."
  ],
  "embedding_hubness": [
    "HEALTHY ITEM DISPERSION: Global Item Uniformity is excellent (-2.83). Movies are well-distributed across the hypersphere.",
    "LOW MOVIE HUBNESS: k-NN occurrence skewness is well-controlled (0.55, Anti-hubs: 0.1%).",
    "USER GEOMETRY HEALTHY: Global User Uniformity (-2.92), Power users (Tier 0: -2.90), and Light users (Tier 2: -2.85) are well-spread across the hypersphere without island collapse.",
    "WEAK ALIGNMENT WARNING: Average positive pair squared distance is 1.70 (Orthogonal/Random is 2.0, Max is 4.0). User embeddings are struggling to align with their ground-truth test items."
  ],
  "all_users_inter_user_diversity_conclusions": [
    "[MODERATE PERSONALIZATION]: Mean Jaccard overlap (5.52%) is 4.2x random chance (1.30%). Candidate slates share core popular hubs while retaining user-specific targeting."
  ],
  "user_tier_0_inter_user_diversity_conclusions": [
    "[MODERATE PERSONALIZATION]: Mean Jaccard overlap (6.26%) is 4.8x random chance (1.30%). Candidate slates share core popular hubs while retaining user-specific targeting."
  ],
  "user_tier_1_inter_user_diversity_conclusions": [
    "[MODERATE PERSONALIZATION]: Mean Jaccard overlap (5.32%) is 4.1x random chance (1.30%). Candidate slates share core popular hubs while retaining user-specific targeting."
  ],
  "user_tier_2_inter_user_diversity_conclusions": [
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    "[MODERATE PERSONALIZATION]: Mean Jaccard overlap (5.39%) is 4.1x random chance (1.30%). Candidate slates share core popular hubs while retaining user-specific targeting."
  ],
  "all_users_and_catalog_has_cold_start_movies_inter_user_diversity_conclusions": [
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  ],
  "intra-list diversity": [
    "[FOCUSED CANDIDATE POOL]: Model ILD (0.4545) retains 47.1% of random catalog diversity. This indicates strong, coherent cluster targeting around user preferences."
  ],
  "user_tier_0_popularity_bias": [
    "[LOG-Q TAIL EXPLORATION]: Retrieved top-100 slates reach deeper into torso/tail inventory (LogPop: 5.75, Ratio: 0.75x, Delta: -1.93) than historical test positives (7.68), counteracting training popularity bias."
  ],
  "user_tier_1_popularity_bias": [
    "[LOG-Q TAIL EXPLORATION]: Retrieved top-100 slates reach deeper into torso/tail inventory (LogPop: 5.42, Ratio: 0.67x, Delta: -2.72) than historical test positives (8.14), counteracting training popularity bias."
  ],
  "user_tier_2_popularity_bias": [
    "[LOG-Q TAIL EXPLORATION]: Retrieved top-100 slates reach deeper into torso/tail inventory (LogPop: 5.47, Ratio: 0.65x, Delta: -2.95) than historical test positives (8.42), counteracting training popularity bias."
  ],
  "cold_start_embeddings": [
    "[SEVERE COLD-START PENALTY]: cold-start set overall recall dropped by 53.2% relative to test set (12.36% -> 5.78%). The item tower overfits to interaction IDs rather than generalizing via metadata.",
    "[TIER 0 COLLAPSE]: Tier 0 recall dropped from 9.01% to 1.24%. Without interaction history, the model's ability to target Tier 0 falls to random chance.",
    "[PRECISION DEFICIT - TIER 0]: Model allocates 21.1% of top-100 candidate slots to Tier 0 items, yet recall (1.24%) is at/below random chance. It retrieves the WRONG items.",
    "[SLATE COMPOSITION - cold-start set]: Top-100 candidate slates consist of 21.1% Tier 0, 42.1% Tier 1, and 36.8% Tier 2 items."
  ],
  "stratified_metrics@20": [
    "STRONG RECALL LIFT: Global Recall@20 (2.4%) heavily outperforms the random expected baseline (0.5%). The candidate generator is extracting meaningful semantic signal.",
    "STRONG PRECISION LIFT: Global Precision@20 (0.88%) heavily outperforms the random expected precision (0.24%), indicating high density of relevant items in the retrieved slates.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 3.3%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 1.4%).",
    "POWER USER RANKING ADVANTAGE: NDCG is higher for power users (Tier 0: 1.60% vs Tier 2: 1.54%).",
```

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      "STRONG LONG-TAIL PERFORMANCE: Model surfaces and ranks niche/long-tail items (Tier 2 NDCG: 1.25%) higher than popular head items (Tier 0 NDCG: 0.99%)."
    ],
    "stratified_metrics@100": [
      "STRONG RECALL LIFT: Global Recall@100 (9.3%) heavily outperforms the random expected baseline (2.6%). The candidate generator is extracting meaningful semantic signal.",
      "MODERATE PRECISION LIFT: Global Precision@100 (0.71%) is better than random (0.24%), suggesting basic relevance filtering is functioning.",
      "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 11.6%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 6.2%).",
      "POWER USER RANKING ADVANTAGE: NDCG is higher for power users (Tier 0: 3.75% vs Tier 2: 3.43%).",
      "STRONG LONG-TAIL PERFORMANCE: Model surfaces and ranks niche/long-tail items (Tier 2 NDCG: 3.32%) higher than popular head items (Tier 0 NDCG: 2.43%)."
    ]
}
```